Few-Shot Incremental Object Detection in Aerial Imagery via Dual-Frequency Prompt

Xiaonan Lu, Wenhui Diao, Junxi Li, Yidan Zhang, Peijin Wang, Xian Sun, *Senior Member, IEEE*, and Kun Fu

***Abstract*— Recently, there has been a growing interest in few-shot incremental object detection (FSIOD). It learns new tasks with limited data while mitigating catastrophic forgetting on previous tasks. However, existing FSIOD methods experience parameter changes after training on new tasks, causing a parameter competition issue among tasks. In addition, the background information differs among various tasks, and using a common background weight for all tasks results in the background shift. Constrained by these two issues, existing methods only alleviate catastrophic forgetting and cannot wholly prevent the performance decline on previous tasks. Especially for complex remote sensing images with messy background, the models trained on new tasks exhibit noticeable performance drops on previous tasks. In this article, we propose a novel FSIOD method via dual-frequency prompt to address these challenges, named FSIOD-DFP. It can completely eliminate catastrophic forgetting while mitigating overfitting. Specifically, a dual-frequency prompt generator is designed to tackle the parameter competition issue. It decouples the frequency components of images to produce prompts that modify the images to adapt to the base model trained on the previous tasks. Compared to traditional prompts, our generator introduces fewer parameters to address overfitting for limited data and allows freezing the base model to maintain the performance of previous data. Besides, a self-regularization loss is introduced to guide the prompt-modified images to leverage the knowledge of the base model effectively. Furthermore, we propose a task-decoupled detection head to address the background shift problem. It separates the detection heads for new and previous tasks to resolve the conflict in the background between different tasks. In FSIOD-DFP, only a prompt generator and a novel detection head are added and fine-tuned when learning a new task. Experiments on three remote sensing object detection datasets demonstrate that our method achieves state-of-the-art performance on both new and previous tasks in all few-shot incremental settings.**

***Index Terms*— Dual-frequency prompt, few-shot incremental learning (FSIL), object detection, remote sensing images.**

I. Introduction

With the development of deep learning, remarkable advancements have been achieved in remote sensing imagery interpretation [1], [2], [3], [4]. However, most deep learning models typically demand abundant data at once for training, following an idealized assumption that all data obeys independent and identically distributed (i.i.d.) [5]. In most remote sensing scenarios, data are often not obtained all at once and may be insufficient due to the scarcity of objects or the difficulty in annotation [6], [7]. Remote sensing images are acquired by sensors with varying loads, spatial resolutions, and spectral ranges. Sequentially incoming remote sensing data exhibit complex and diverse distribution changes, deviating from the i.i.d. assumption. When learning on newly incoming data, most deep learning models meet catastrophic forgetting [8]. Also, when the amount of sequentially incoming data is limited, the models trained on few-shot samples also face overfitting [9], failing to achieve optimal performance.

In response to the challenges, several incremental learning approaches [10], [11], [12] have been designed to learn sequentially incoming data and alleviate catastrophic forgetting when learning new tasks. Also, there are some few-shot methods [13], [14], [15], [16], [17], [18], [19], [20] aimed at enhancing model's generalization and addressing overfitting in the situations where training data are limited. However, as shown in Fig. 1, the former assumes that the training data of each task are abundant and fail to address overfitting when data are limited. The latter focuses solely on the performance of newly incoming limited data, neglecting the catastrophic forgetting on previous data. Moreover, there are a few methods that integrate the two kinds of approaches, namely, few-shot incremental learning (FSIL) [21], to simultaneously address the catastrophic forgetting issue induced by sequentially incoming data and the overfitting issue caused by limited data. It is especially crucial in real-world applications where acquiring sufficient labeled data is impractical or cost-prohibitive.

FSIL assumes that an initial task containing base classes with abundant data is available for training to acquire a base model. Subsequently, the incoming training data for new tasks are limited. FSIL expects the model to avoid overfitting for

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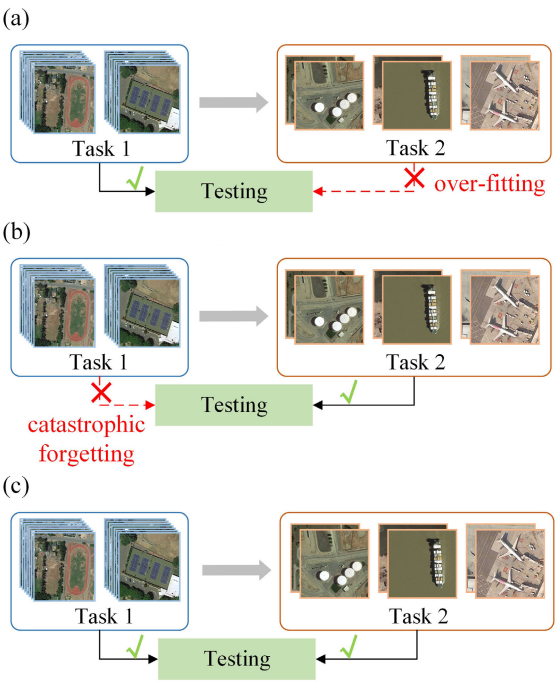


Fig. 1. Schematic of incremental learning, few-shot learning, and FSIL. (a) Incremental learning addresses catastrophic forgetting for sequentially arriving data, but it may suffer from overfitting for limited data. (b) Few-shot learning can alleviate overfitting but tends to forget previous knowledge after learning new tasks. (c) FSIL, combining the two approaches, can concurrently mitigate catastrophic forgetting and overfitting.

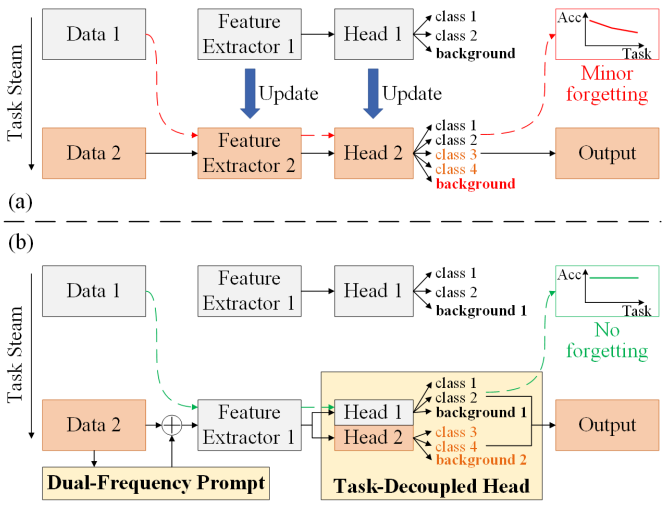


Fig. 2. Comparison of FSIOD methods. (a) Due to parameter competition and background shift, previous works could only alleviate catastrophic forgetting and still result in minor forgetting on previous tasks. (b) Our method, introducing dual-frequency prompt and task-decoupled head, can completely eliminate catastrophic forgetting while mitigating overfitting, without experiencing a performance decline on previous tasks.

new tasks while retaining the performance on previous tasks after training on new data. While some FSIL methods have achieved success in classification task [22], [23], [24], [25], there is a scarcity of approaches for few-shot incremental object detection (FSIOD), which is more complex than FSIL. Existing FSIOD methods [26], [27] usually modify the model structure or impose constraints on parameters to make the model adaptable to FSIOD scenarios. However, as shown in Fig. 2(a), these methods only mitigate catastrophic forgetting, meaning that the performance on previous tasks still experiences a decline after training on new data, which is more pronounced in remote sensing imagery, primarily due to two issues.

1) *Parameter Competition:* Both new and old tasks share some common parameters. It leads to changes in some parameters crucial for previous tasks when learning new tasks, causing catastrophic forgetting. Also, compared to natural scene images, there is a vast diversity between remote sensing images. For remote sensing, training on new data will cause more noticeable parameter changes, resulting in a severe decline in performance on previous data.
2) *Background Shift:* Most methods employ a single detection head for all tasks, which encounters the issue of background shift. The foreground objects of the new task are treated as background during learning previous tasks, and the objects of previous tasks are also perceived as background when learning the new task. It results in the inability of models to accurately distinguish foreground objects and background, especially for remote sensing images with complex background.

In pursuit of a method that can completely eliminate catastrophic forgetting while alleviating overfitting, ensuring that the model trained on new data does not experience performance decline on previous tasks, as shown in Fig. 2(b), we propose a novel FSIOD approach via dual-frequency prompt, called FSIOD-DFP.

First, for the parameter competition issue, recent advancements in prompt learning [28], [29] for the fine-tuning domain have inspired our approach. Prompt learning enhances the performance on downstream tasks by modifying input data under a frozen pretrained model. Prompts typically involve learnable parameters added to or concatenated with the original data before inputting into the model. However, traditional prompts are designed for downstream tasks with plenty of data. Directly applying them to FSIOD will meet overfitting due to introducing many training parameters. Therefore, we design a dual-frequency prompt generator based on traditional prompts to extract prompts for learning new tasks. The generated prompts modify the input images to adapt to the existing model that performs well on previous data. With the advancement of remote sensing and super-resolution technologies [30], [31], [32], [33], the resolution of remote sensing images is gradually improving. High-resolution remote sensing images encompass a wealth of geographic information, such as fine textures and intricate details, leading to significant interference between adjacent geographical features. Wang et al. [34] demonstrated from a data perspective that convolutional neural networks exhibit different inductive biases for low- and high-frequency components. Among them, the low-frequency component tends to be more generalizable and contains essential information about the main structure, while the high-frequency component captures boundary details and allows for effective differentiation between various objects, potentially playing a significant role in depicting objects. Compared to the channel domain, the frequency domain exhibits higher distinguishability between different frequency components, making it easier to filter out interference information. Thus, based on the fast

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Fourier transform (FFT) and filtering techniques, our proposed prompt generator decouples low- and high-frequency information and applies a dual-branch fusion structure for prompt generation. It effectively filters out background interference and notably decreases the number of introduced training parameters for mitigating overfitting. Besides, to effectively utilize the general information from the base model trained on abundant data, we introduce a self-regularization loss for explicitly guiding the training trajectory through maximizing mutual agreement between the prompt-modified images and the frozen base model.

Furthermore, a task-decoupled detection head is proposed to address the background shift issue. The detection head of each task possesses independent background weights and does not affect each other, avoiding background confusion between different tasks. Also, to alleviate intraclass differences caused by limited data for new tasks, we employ a cosine similarity (CS) classifier [18] for new tasks. It disregards the magnitude of features through normalization and solely focuses on the direction of features, which enhances the generalization for new classes.

The efficacy of our proposed FSIOD-DFP is validated on three remote sensing object detection datasets, including DIOR [35], NWPU VHR-10 [36], and FAIR1M [37]. The experimental results conducted on individual datasets and a sequence of multiple datasets illustrate that the proposed FSIOD-DFP achieves state-of-the-art performance in all settings.

The contributions of this article are summarized as follows.

1) We propose an FSIOD approach for remote sensing images, called FSIOD-DFP. It not only mitigates overfitting but also completely eliminates catastrophic forgetting. Besides, it does not require replaying the data of previous tasks while learning new tasks and is applicable across various object detection models.
2) We introduce a dual-frequency prompt generator to address catastrophic forgetting caused by parameter competition. It can also alleviate overfitting compared to conventional prompts. Also, a self-regularization loss is proposed to utilize the knowledge of the base model effectively.
3) A task-decoupled detection head is proposed to resolve the background shift problem through an independent detection head for each task. Also, a CS classifier is designed to enhance the generalization for new tasks with limited data.
4) Our FSIOD-DFP achieves state-of-the-art performance on three datasets and demonstrates optimal performance in complex scenarios across multiple datasets.

II. Related Works

In this section, we review pertinent literature on FSIL, prompt learning, and object detection, particularly within the realms of computer vision and remote sensing.

A. Few-Shot Incremental Learning

With the development of few-shot learning [38] and incremental learning [39], FSIL has garnered significant attention recently. It first trains a base model on a dataset with abundant data for base classes and fine-tunes the base model on sequentially incoming limited data for novel classes, demanding efficient adaptation to new data without experiencing knowledge forgetting for previous data. It can broadly be categorized into fine-tuning with constraint methods, parameter selection update methods, and decoupled learning methods. The fine-tuning with constraint methods [21], [22] fine-tune network parameters for learning new tasks while mitigating catastrophic forgetting caused by parameter changes through extra constraint losses. Tao et al. [21] designed a neural gas network to preserve the topology of the feature manifold and propose a topology-preserving knowledge incrementer framework aimed at mitigating the forgetting of old classes. Cheraghian et al. [22] introduced a distilled FSIL algorithm. It applies class words as semantic information to align visual features and semantic features. The parameter selection update methods [40] employ strategies to select parameters that are not crucial for previous tasks and only update the selected parameters during fine-tuning for new tasks. In few-shot lifelong learning (FSLL) [40], only a few parameters that are unimportant for previous tasks get selected, and the CS between the prototypes of old and new classes is introduced to make them separated and improve the classification accuracy. The decoupled learning methods [23], [24], [25] decouple the feature extractor and the head network, allowing independent training of the head network when learning new tasks. Continually evolved classifier [24] utilizes a graph model and a pseudo incremental learning paradigm to propagate information between classifiers for adaptation. Continual prototype calibration [7] addresses the challenges posed by remote sensing images by employing the decoupled learning strategy and calibrating interclass and intraclass prototypes.

Most FSIL models are designed for classification tasks, with only limited methods proposed for FSIOD [26], [27], [41], [42]. ONCE [26] introduces an adaptation strategy to CentreNet detector [43] and designs a class-specific code generator tailored explicitly for new categories. iMTFA [41] trains discriminative embeddings for objects, which are then merged into class features by using CS. Incremental-DETR [42] achieves FSIOD via knowledge distillation on class-specific components and self-supervised training on DETR [44]. G-FSDet [27] designs an efficient transfer-learning model for generalized few-shot object detection in remote sensing images.

These models primarily mitigate catastrophic forgetting on previous tasks but fall short of completely eliminating forgetting. However, our approach is capable of learning new tasks without encountering performance decline on previous tasks.

B. Prompt Learning

Prompt learning has initially gained widespread application in natural language processing, especially for large-scale model fine-tuning [28], [29]. It guides models in performing specific tasks by refining input data through well-designed prompts. With the development of large-scale visual models [45], [46], [47], some studies have introduced prompt

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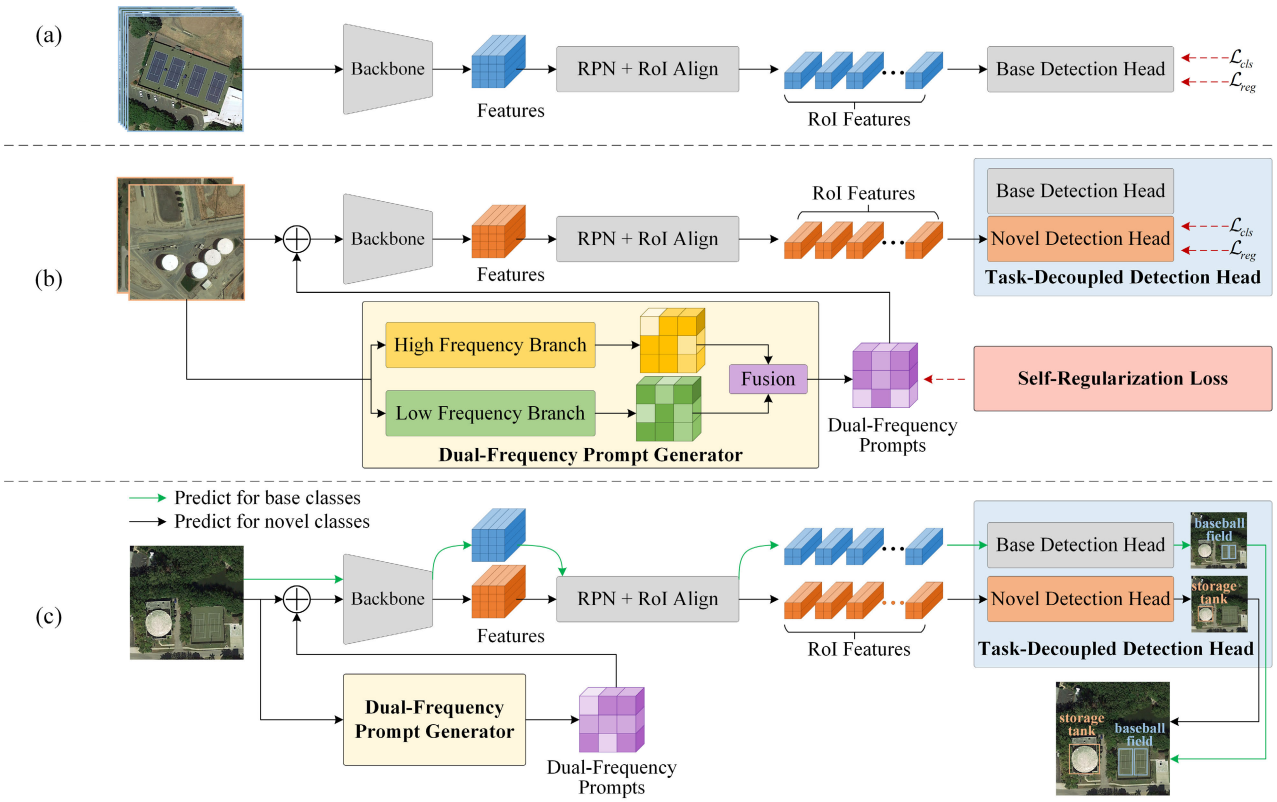


Fig. 3. Architecture and pipeline of our FSIOD-DFP. (a) In base training, an object detector is trained on a base dataset with abundant data to acquire a base model that excels on base classes. (b) Subsequent training is employed for new tasks with limited data. All parameters of the base model are frozen to maintain the performance on previous tasks. A dual-frequency prompt generator and a task-decoupled detection head are introduced. The generated prompts adapt the input data to the base model for predicting objects for new classes. (c) During testing, the original input image is first fed into the backbone, RPN, RoI Align, and the base detection head to predict objects of base classes, indicated by the green arrow. Then, the prompt-modified image is sent to the backbone and the novel detection head to predict objects of novel classes, indicated by the black arrow. All predictions are integrated as the final output.

learning into computer vision [48], [49], [50]. Visual prompts are typically learnable parameters added to or concatenated with the original data used as the input of models. DenseCLIP [51] uses the contextual features from images to prompt the large-scale language model for dense prediction. Explicit visual prompting [52] enforces the parameters focusing on definite contents on image-level rather than dataset-level implicit embeddings. Shtedritski et al. [53] simply draw a red circle around an object as visual prompts for solving computer vision tasks. In diversity-aware meta visual prompting [54], the prompts are initialized with a module learned on multiple datasets. It helps the prompts accelerate convergence and improve the performance on new datasets.

Currently, there are also several works focused on designing visual prompts for few-shot learning and incremental learning [55], [56], [57]. For few-shot learning, ProD [58] applies the prompting mechanism to decouple the domain general and specific features for few-shot image classification. FS-DETR [59] introduces templates of novel classes as visual prompts for few-shot object detection. In addition, for incremental learning, DualPrompt [60] trains a prompt set to effectively guide a pretrained model in learning sequential tasks without the need to buffer past examples. S-Prompting [61] independently learns prompts among domains with pretrained transformers to address catastrophic forgetting. Gan et al. [62] propose a continual test-time adaptation method to train an image-level prompt for the target domain and freeze the source model, which adapts the source model to continually changed target domains.

Existing visual prompt methods are typically applied to downstream tasks with large datasets. Directly applying them to FSIOD scenarios will introduce overfitting due to the introduction of excessive trainable parameters. Thus, we design a dual-frequency prompt generator to reduce the number of trainable parameters for learning new tasks.

C. Object Detection

Object detection holds pivotal importance across various applications, such as autonomous vehicles, urban planning, and remote sensing surveillance systems. The common detectors can be categorized into two- and single-stage methods. Two-stage object detectors [63], [64], [65] extract candidate regions of interest (RoIs) followed by classification and regression. In contrast, single-stage methods [66], [67], [68], [69] eliminate the proposal extraction step and directly leverage a deep learning neural network to predict object labels and locations. With the continuous development of object detectors and remote sensing data fusion [70], [71], the accessibility of high-resolution remote sensing images has propelled advancements in remote sensing object detection [1], [2], [3], [4]. FMSSD [1] unifies contextual information by incorporating multiple atrous convolutional layers for small objects. R3Det [2] specializes in detecting objects with rotating

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attributes, substantial aspect ratios, and dense arrangements. CODet [4] comprises a cross-hierarchy feature fusion model to fuse multiple resolution features and a noise-sparse sample assignment method for detecting component objects in remote sensing images with complex characteristics. However, most object detection methods, while successful in scenarios with abundant labeled data, face considerable challenges when confronted with limited and sequentially incoming data. We propose an FSIOD approach to enhance the performance and adaptability in resource-constrained environments.

III. Methodology

In this section, we provide a detailed description of our FSIOD-DFP. First, we introduce the problem setup of FSIOD. Then, the overall framework of our proposed FSIOD-DFP is outlined. Subsequently, we separately provide the detailed descriptions of the designed components, including dual-frequency prompt generator, self-regularization loss, and task-decoupled detection head.

A. Problem Setup

FSIOD aims to address sequentially arriving limited data and simultaneously alleviate catastrophic forgetting and overfitting when training on new limited data. In other words, it expects that the model, without accessing previous data, can achieve commendable performance on both old and new tasks by learning solely on the limited new data.

Specifically, assuming that there are T tasks, let us denote $\{D^0_{\text{train}}, D^1_{\text{train}}, \ldots, D^{T-1}_{\text{train}}\}$ as training datasets of different tasks for FSIOD. For the ith task, C^i represents the specific categories that should be learned. The number of categories for few-shot incremental tasks is not fixed, and FSIOD assumes that categories between different tasks are nonoverlapping, i.e., $C^i \cap C^j = \oslash$ for $i \neq j$. In task i, the training dataset $D^i_{\text{train}} = \{x^i_j, y^i_j\}^{N^i_{\text{train}}}_{j=1}$, where x^i_j represents the jth image of task i with objects of C^i and y^i_j represents the objects labels and bounding boxes. N^i_{train} means the number of labeled images of task i. It is noteworthy that the first dataset D^0_{train} is called the base dataset with base categories. Each base category comprises a substantial amount of annotated data. The model trained on D^0_{train} is referred to a base model, serving as the foundation for subsequent training. For the subsequent datasets $D^i_{\text{train}}(i > 0)$, there are only limited data for each category, where k-shot setting indicates that each category has k annotated instances.

When learning the ith task, the model only accesses the dataset D^i_{train} for training and cannot replay the data of previously learned tasks. After learning the ith task, the model is evaluated on the test set $\{D^0_{\text{test}}, D^1_{\text{test}}, \ldots, D^i_{\text{test}}\}$, which contains test images in both previous and current tasks with classes $C^0 \cup C^1 \cup \cdots \cup C^i$. The FSIOD methods not only need to overcome overfitting caused by limited data for the current task but also to mitigate catastrophic forgetting for previously learned tasks, which present distinguished performance on both current and previous tasks. Our method can completely eliminate catastrophic forgetting caused by parameter competition and background shift by introducing prompts and task-decoupled detection head, ensuring that the knowledge learned from previous tasks is not forgotten when learning new tasks.

B. Overview

To address the minor forgetting in existing FSIOD methods caused by parameter competition and background shift, we propose a novel FSIOD approach based on dual-frequency prompt, called FSIOD-DFP. It can completely eliminate catastrophic forgetting while mitigating overfitting, ensuring that the performance on previous tasks does not decline after training on new tasks. Also, our model is adaptable to various object detection models. Taking Faster RCNN [63] as an example, the overall architecture and pipeline of FSIOD-DFP are depicted in Fig. 3. It primarily consists of a dual-frequency prompt generator, a self-regularization loss, and a task-decoupled detection head. Also, the process of our model consists of base training, subsequent training, and testing.

FSIOD does not require models to learn few-shot samples from scratch. Instead, it expects models to fully utilize the knowledge learned from a rich dataset with base classes and apply the knowledge to novel classes with limited samples, enhancing the performance on novel classes and maintaining the performance on base classes. Thus, our method consists of two training stages: base training and subsequent training. First, in the base training stage, a conventional object detection model is trained on the base dataset D^0_{train}, which contains abundant data with base classes. After base training, we acquire a base model with excellent performance on base classes, which is used for subsequent training. During base training, all parameters of the detector are trained, and the conventional detection loss is applied, including classification loss $\mathcal{L}_{\text{cls}}$ and regression loss $\mathcal{L}_{\text{reg}}$, which is described as

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} \tag{1}$$

where cross-entropy loss and $L1$ loss are applied as the classification loss and the regression loss, respectively.

Then, when a new task containing new classes with limited data arrives, the base model needs subsequent training. Based on the base model, we introduce a dual-frequency prompt generator in the subsequent training stage to generate prompts for modifying input images. Also, a novel detection head is added to predict objects belonging to new classes. This task-decoupled form of the novel and base detection heads addresses the background shift issue. During subsequent training, only the parameters of the newly added components are trained, and the base model is frozen to ensure that the performance on previous tasks does not degrade. The dual-frequency prompt generator decouples the image into low- and high-frequency components to generate dual-frequency prompts. They modify the distribution of input images to adapt them to the frozen base model, allowing accurate feature extraction and candidate box prediction. The extracted RoI features are then fed into the novel detection head of the task-decoupled detection head to detect objects of new classes. Furthermore, to explicitly guide the generated prompts to modify the input data effectively and utilize the general knowledge of the base model, we propose a

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Algorithm 1 Pipeline of FSIOD-DFP

Input: Base dataset D^0_{train} and incremental few-shot datasets $\{D^1_{train}, D^2_{train}, \ldots, D^{T-1}_{train}\}$. The initial parameters of the base detector θ^0 and the learning rate lr.

Output: The parameters of the model after training on all tasks θ^{T-1}. The predicted results of the test image res.

Stage 1: Base Training
Forward propagating the image x of base dataset D^0_{train} on the base model to predict classes and locations of objects belonging to base classes
Calculate the base training loss $\mathcal{L}_{base}$ in Eq. (1)
Backward propagation and update $\theta^0 \leftarrow \theta^0 - lr\frac{\partial \mathcal{L}_{base}}{\partial \theta^0}$
Stage 2: Subsequent Training
for i in $[1, 2, \ldots, T-1]$ **do**
    Load the weights θ^{i-1} of the model trained on Task $i-1$ and freeze them
    Add a dual-frequency prompt generator and a novel detection head to the model for Task i
    Forward propagating the image x of dataset D^i_{train} to the prompt generator to obtain prompts
    Modify x with prompts and send them to the backbone and the novel detection head to predict objects belonging to the classes of Task i
    Calculate the training loss $\mathcal{L}_{subsequent}$ in Eq. (2)
    Backward and update $\theta^i \leftarrow \theta^i - lr\frac{\partial \mathcal{L}_{subsequent}}{\partial \theta^i}$
end for
return θ^{T-1}
Stage 3: Testing
Load θ^{T-1} of the model and give a test image x
Forward propagating x to the backbone and the base detection head to predict objects belonging to the base task, denoted as res^0
for i in $[1, 2 \ldots, T-1]$ **do**
    Forward propagating x to the prompt generator corresponding to Task i to obtain prompts
    Modify x with the generated prompts and send them to the backbone and the novel detection head of Task i to predict objects belonging to Task i, denoted as res^i
end for
Integrate all predicted objects as the final result of the test image, i.e., $res = \{res^0, res^1, res^2, \ldots, res^{T-1}\}$
return res

self-regularization loss $\mathcal{L}_{\text{self}}$ along with the object detection loss for subsequent training, which is defined as follows:

$$\mathcal{L}_{\text{subsequent}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \lambda \mathcal{L}_{\text{self}} \tag{2}$$

where λ is the balance weight between the self-regularization loss and the object detection loss. Whenever a new task with limited data arrives, a new dual-frequency prompt generator and a novel detection head are added, followed by the subsequent training process. Note that unlike previous methods [17], [20], [27], our approach does not require accessing the data of previous tasks during the subsequent training stage. It can completely maintain the performance of previous tasks without experiencing catastrophic forgetting.

After training on all tasks, the testing process is shown in Fig. 3(c). Given an input image, it is first directly fed into the backbone and the base detection head to predict objects belonging to base classes, indicated by the green arrow. Then, as the black arrow shows, the image is input into the dual-frequency prompt generator to obtain prompts. The generator applies low- and high-pass filtering to the input image to filter out interference and then uses a dual-branch fusion structure to simultaneously extract the objects' main body information from the low-frequency component and the edge detail information from the high-frequency component to generate prompts. The prompts are added with the original input image and then sent into the backbone and the novel detection head to predict objects belonging to new classes. If there are multiple subsequent tasks, the input image is sequentially processed by the dual-frequency prompt generator and the novel detection head corresponding to each task to obtain predictions for the respective categories. Finally, all predictions are integrated as the final output, encompassing objects with all classes. The pipeline of FSIOD-DFP can be seen in Algorithm 1.

C. Dual-Frequency Prompt Generator

We draw inspiration from prompt learning to address the catastrophic forgetting caused by parameter competition. By freezing the model and introducing prompts to modify input data, it can adapt the input to the model and maintain the performance on previous tasks when learning new tasks. Conventional prompt learning typically introduces learnable embeddings that are added or concatenated with input data, followed by feeding them into the model for feature extraction and prediction. It is commonly applied in fine-tuning downstream tasks with abundant data. However, due to the excessive number of introduced parameters, directly applying it to tasks with limited data may lead to overfitting.

Thus, by combining the presentation content and frequency characteristics of optical remote sensing images, we design a dual-frequency prompt generator to reduce the number of learnable parameters for prompt generation. Remote sensing images often encompass vast scenes with intricate and varied background textures. In scenarios with limited data, the model may struggle to extract features of objects from cluttered background accurately. As shown in Fig. 4(a), if the image is filtered to separate low- and high-frequency components, the low-frequency component will filter out background textures, such as grass, while retaining the main structure of objects. Also, the high-frequency component will preserve the boundary information. Hence, decomposing the input image into low- and high-frequency components and using both of them are conducive to generating effective prompts with fewer parameters.

The dual-frequency prompt generator first performs frequency filtering on the input image to obtain dual-frequency components. We utilize the FFT and inverse FFT (iFFT) to achieve this function. For an input image I with sizes of $H \times W$, let z represent its frequency spectrum, i.e., $z = \text{FFT}(I)$. Then, the frequency coefficients are shifted to the center of the image $((H/2), (W/2))$. For high- and low-pass filtering,

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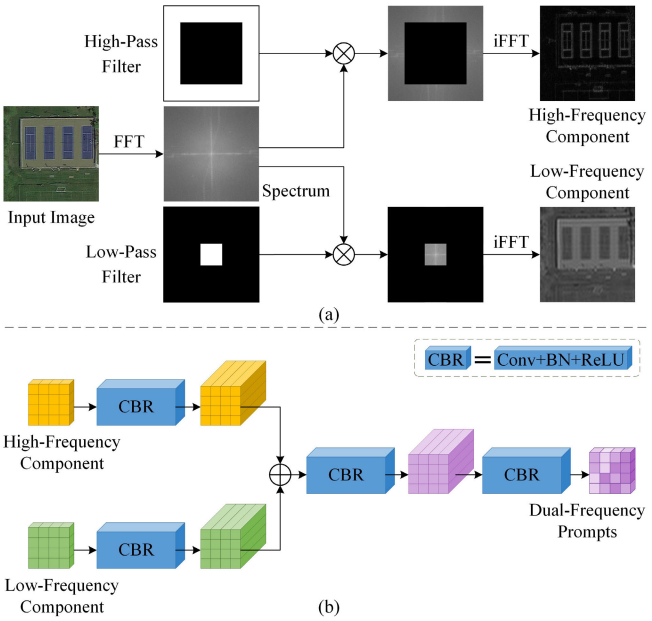


Fig. 4. (a) Procedure to generate high- and low-frequency components from the input image. (b) Structure of dual-frequency prompt generator.

the corresponding binary masks M_h and M_l are generated, respectively, which can be presented as follows:

$$M_h^{i,j} = \begin{cases} 0, & \left|i - \frac{H}{2}\right| \le \alpha_h H \text{ and } \left|j - \frac{W}{2}\right| \le \alpha_h W \\ 1, & \text{otherwise} \end{cases} \tag{3}$$

$$M_l^{i,j} = \begin{cases} 1, & \left|i - \frac{H}{2}\right| \le \alpha_l H \text{ and } \left|j - \frac{W}{2}\right| \le \alpha_l W \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where α_h and α_l represent the surface ratio of the centered regions for the high-pass filter and low-pass filter, respectively. Then, the high- and low-frequency components are calculated as follows:

$$I_h = \text{iFFT}(zM_h) \tag{5}$$

$$I_l = \text{iFFT}(zM_l). \tag{6}$$

For RGB images, the process is computed on each channel independently. After obtaining the dual-frequency components, we use a few convolutional layers to generate prompts, as shown in Fig. 4(b). First, two 1×1 convolutional layers are utilized to produce features from each component and increase their dimensions. Subsequently, the features are fused through addition, followed by a 3×3 convolutional layer to generate fused features. Finally, a 1×1 convolutional layer is utilized to convert the dimension to three channels, yielding the dual-frequency prompts for adding to the original input image. The process can be represented as follows:

$$F_h = \text{Conv}_h(I_h), \quad F_l = \text{Conv}_l(I_l) \tag{7}$$

$$\text{DFP} = \text{Conv}_{1\times1}(\text{Conv}_{3\times3}(F_h + F_l)) \tag{8}$$

where DFP means the dual-frequency prompts. The proposed generator effectively utilizes the information of the objects' main body from the low-frequency component and the boundary information from the high-frequency component, merging them to create representative prompts for new tasks. To adapt to different networks, we use an addition operation to fuse the generated dual-frequency prompts and the input image, modifying the image to adapt to the base model.

Regarding the number of trainable parameters, traditional prompts require a learnable embedding with the same size as the input image for fine-tuning, with a parameter count of $3HW$, while our proposed dual-frequency prompt generator only consists of four convolutional layers and four BatchNorm layers. The convolutional layers are with parameter counts of $1 \times 1 \times 3 \times d$, $1 \times 1 \times 3 \times d$, $3 \times 3 \times d \times d$, and $1 \times 1 \times d \times 3$, where d means the increased dimension of the features, which is far smaller than H and W. Also, the BatchNorm layers are with parameter counts of $2d$, $2d$, $2d$, and 6. Thus, the number of introduced parameters in our generator is summing up to $(9d + 6)(d + 1)$. Taking the DIOR dataset used in our experiments as an example, H and W are both 800, and d is 64. The number of parameters for traditional prompt learning is 1 920 000, while our proposed generator has only 37 830 parameters, accounting for just 1.97% of the conventional prompt. Therefore, the number of training parameters in our proposed dual-frequency prompt generator is significantly lower than that of conventional prompt learning, which can effectively alleviate overfitting for FSIOD.

D. Self-Regularization Loss

In addition to the conventional detection loss, we explicitly introduce a loss during subsequent training to ensure that the generated prompts can fully leverage the knowledge of the base model trained on the extensive base dataset. It aims to maximize the consistency between the features extracted from the prompt-modified images and the features extracted from the original input images. Since there is no need for an extra model, we call it as self-regularization loss.

Given an input image, it passes through the backbone of the base model directly to obtain the features denoted as f. Then, the image goes through the dual-frequency prompt generator to generate prompts, which are added to the input image and then fed into the backbone of the base model to obtain features denoted as $\widetilde{f}$. The self-regularization loss $\mathcal{L}_{\text{self}}$ applies the $L1$ loss to enforce consistency between the two features, which is calculated as follows:

$$\mathcal{L}_{\text{self}} = \left| f - \widetilde{f} \right|. \tag{9}$$

The self-regularization loss can also be formulated with other forms, such as CS and mean squared error (mse) loss. The total loss during the subsequent training stage is shown in (2). Therefore, the object detection loss is used to train the dual-frequency prompt generator and the novel detection head to achieve optimal performance on the current task by imposing explicit constraints on the model with ground truth annotations. The self-regularization loss guides the learning trajectory of the generated prompts toward a weight space consistent with the frozen base model, thereby leveraging the generalization knowledge of the base model.

E. Task-Decoupled Detection Head

The previous tasks treat the objects of the new task as background during previous training, and the new task also

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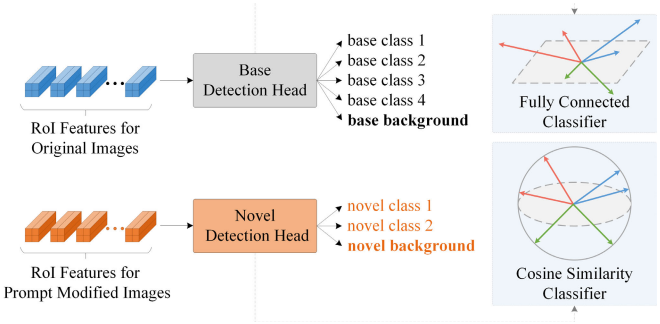


Fig. 5. Structure of the task-decoupled detection head.

treats the foreground objects of previous tasks as background due to the lack of annotations. The information pointed to the background in different tasks is not the same. It is necessary to completely decouple the detection heads of different tasks to address the background shift problem. Thus, a task-decoupled detection head is proposed, as shown in Fig. 5. It completely decouples the detection heads of the novel and old tasks. The background weights are independent between different tasks.

For the novel detection head in the proposed task-decoupled detection head, the regression head uses a fully connected (FC) layer with random initialization, while the classification head, same as TFA [18] and G-FSDet [27], employs a CS classifier, which is represented as follows:

$$s_{i,j} = \frac{\alpha F(x)_i w_j}{\|F(x)_i\| \|w_j\|} \tag{10}$$

$$p_{i,j} = \frac{e^{s_{i,j}}}{\sum_{j=0}^{C} e^{s_{i,j}}} \tag{11}$$

where $F(x)_i$ means the ith RoI features and w_j is the classification weight for the jth class. $\alpha = 20$ is the scale factor. $s_{i,j}$ and $p_{i,j}$ are the CS score and the final predicted score that the ith RoI features belong to the jth class, respectively. Compared to the FC classifier, the CS classifier normalizes the features and the classification weights, which reduces the massive intraclass variances caused by limited data and enhances the model's generalization.

When learning a new task, the task-decoupled detection head simply adds a novel detection head. The detection heads of previous tasks are frozen to maintain their performance, and only the novel detection head of the current task is trained. During testing, the features extracted from the original input image and the image modified with prompts generated by each task's prompt generator are separately sent into the base detection head and the novel detection heads corresponding to each task, predicting objects belonging to the classes of the respective task. Finally, the predictions of all tasks are aggregated to obtain the final results for the input image, encompassing the categories of all tasks.

IV. Experiments

In this section, we initially detail the remote sensing object detection datasets and novel class split settings for FSIOD. Subsequently, we outline the evaluation metrics and implementation details. Finally, we demonstrate the effectiveness of the proposed method through extensive ablation studies and comparisons with state-of-the-art methods.

A. Datasets

1) DIOR [35]: The DIOR dataset serves as a comprehensive object detection benchmark for remote sensing images. It encompasses 23 463 images sourced from the Google Earth platform, containing 192 472 object instances. The dataset is systematically divided into training set, validation set, and test set, comprising 5862, 5863, and 11 738 images, respectively. The images maintain a fixed size of 800 × 800 pixels, and their spatial resolutions span from 0.5 to 30 m. The objects in the DIOR dataset are labeled into 20 categories, including airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. Following the few-shot object detection setup in [17] and [27], the dataset adopts four distinct novel/base class splits. Each split comprises 15 base classes and five novel classes. The novel classes are delineated in Table I, and the remaining classes are base classes. For the k-shot setting, k instance annotations are randomly selected from the training and validation sets for each class. Also, k is set to 3, 5, 10, and 20. For testing, the method is systematically evaluated on the test set with both base classes and novel classes.

2) NWPU VHR-10 [36]: The NWPU VHR-10 dataset comprises 715 high-resolution remote sensing images sourced from Google Earth and 85 pansharpened color infrared images of the Vaihingen dataset [72]. The NWPU VHR-10 dataset has 650 positive images featuring 3775 object instances and 150 negative images without foreground objects. The NWPU VHR-10 dataset encompasses ten classes, including airplane, baseball diamond, basketball court, bridge, harbor, ground track field, ship, storage tank, tennis court, and vehicle. To facilitate FSIOD settings, as shown in Table I, we follow the novel classes splits introduced in [73] and [27], with seven base classes and three novel classes. Also, k is also set to 3, 5, 10, and 20. The performance evaluation encompasses all classes within the test set.

3) FAIR1M [37]: The FAIR1M dataset serves as a benchmark specifically designed for fine-grained object detection of remote sensing images, exceeding 15 000 images and featuring over 1.02 million object instances. The spatial resolution varies between 0.3 and 0.8 m, and the size of images spans from 1000 to 10 000 pixels. The annotated objects are meticulously categorized into five overarching classes, further branching into 37 subcategories. To evaluate the efficacy of FSIOD-DFP in the fine-grained object detection scenario, following [20], we conduct experiments on the airplane category, denoted as FAIR1M-Airplane. It contains 11 subcategories with Boeing 737, Boeing 747, Boeing 777, Boeing 787, Airbus A220, Airbus A321, Airbus A330, Airbus A350, COMAC C919, COMAC ARJ21, and other airplane. As shown in Table I, we follow [20] that selects three classes as novel classes and the other eight classes as base classes. The rotation annotations are transformed into horizontal annotations, as we specifically target in FSIOD rather than rotated object detection. Also, k is set to 2, 3, 5, and 10.

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TABLE I
DIFFERENT NOVEL CLASS SPLIT SETTINGS FOR FSIOD SCENARIOS

Dataset	Split	Novel Classes				
DIOR	1	Baseball field	Basketball court	Bridge	Chimney	Ship
	2	Airplane	Airport	Expressway toll station	Harbor	Ground track field
	3	Dam	Golf course	Storage tank	Tennis court	Vehicle
	4	Express service area	Overpass	Stadium	Train station	Windmill
NWPU VHR-10	1	Airplane	Baseball diamond	Tennis court		
	2	Basketball court	Ground track field	Vehicle		
FAIR1M-Airplane	1	A350	Boeing737	Boeing747		

TABLE II
ABLATION STUDY FOR THE DESIGNED COMPONENTS

Components			5-shot			10-shot		
DFPG	$\mathcal{L}_{self}$	TDDH	Base	Novel	All	Base	Novel	All
			69.95	13.23	55.77	70.14	16.15	56.64
✓			67.35	26.83	57.22	67.79	32.96	59.08
		✓	**71.59**	19.53	58.58	**71.59**	25.20	59.99
✓		✓	**71.59**	29.52	61.07	**71.59**	36.33	62.78
✓	✓		69.30	27.97	58.97	69.68	35.83	61.22
✓	✓	✓	**71.59**	**31.34**	**61.53**	**71.59**	**37.98**	**63.19**

* Abbreviation of components - DFPG: Dual-frequency prompt generator; TDDH: Task-decoupled detection head.

TABLE III
ABLATION STUDY FOR DIFFERENT BACKBONES AND DETECTORS

Methods	Backbone	5-shot			10-shot		
		Base	Novel	All	Base	Novel	All
Faster RCNN w DFP	ResNet-50	69.87	31.10	60.18	69.87	36.59	61.55
	ResNet-101	71.59	31.34	61.53	71.59	37.98	63.19
	Swin-Tiny	72.78	31.82	62.54	72.78	38.25	64.14
RetinaNet	ResNet-101	70.01	12.68	55.68	70.62	16.10	56.99
RetinaNet w DFP	ResNet-101	71.65	31.59	61.64	71.65	37.92	63.22
CornerNet	Hourglass-104	68.76	9.35	53.91	69.13	13.97	55.34
CornerNet w DFP	Hourglass-104	70.29	29.54	60.10	70.29	36.15	61.76

* w DFP: with DFP, incorporating the proposed DFP method, including dual-frequency prompt generator, self-regularization loss, and task-decoupled detection head, into conventional object detectors.

TABLE IV
ABLATION STUDY FOR PROMPT GENERATION METHODS

Prompt Methods	5-shot			10-shot		
	Base	Novel	All	Base	Novel	All
No Prompt	**69.95**	13.23	55.77	**70.14**	16.15	56.64
Conventional Prompt	69.89	12.25	56.23	69.95	14.33	56.05
DFPG w Original Image	66.27	20.56	54.84	66.35	23.86	55.73
DFPG w LFC	67.28	24.91	56.69	67.49	31.05	58.38
DFPG w HFC	67.25	23.25	56.25	67.73	28.96	58.04
DFPG w LFC & HFC	67.35	**26.83**	**57.22**	67.79	**32.96**	**59.08**

* w X: with X, the prompt generator use X to produce prompt vectors.
* Abbreviation of prompt methods - LFC: Low-Frequency Component; HFC: High-Frequency Component.

B. Evaluation Metrics

To verify the FSIOD performance, we adopt the same evaluation metrics as previous works [17], [20], [27] which utilize the mean average precision (mAP) with an intersection over union (IoU) threshold of 0.5. Initially, we calculate the number of samples with true positive (TP), false positive (FP), and false negative (FN) based on the IoU between the predicted results and the ground truth. Then, we compute precision (P) and recall (R) as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \tag{12}$$

Ultimately, by plotting the precision–recall curve for each category and calculating the area from 0 to 1 under the curve as the average precision (AP), this can be represented as follows:

$$\mathrm{AP} = \int_0^1 P dR. \tag{13}$$

The mAP is calculated as the average of AP across classes. Unlike few-shot studies [17], [20] that exclusively assess the performance of novel classes, our work also emphasizes the performance of base classes. Thus, similar to [27], we select the mAP of base classes, novel classes, and all classes as FSIOD evaluation metrics.

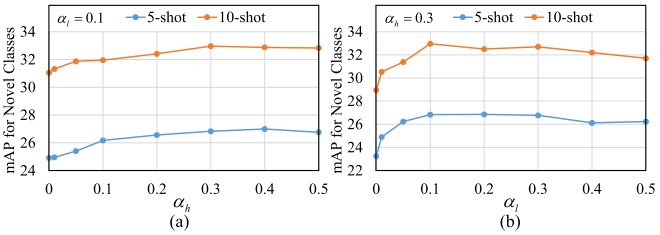


Fig. 6. (a) Ablation study for the surface ratio α_h of the high-pass filter. (b) Ablation study for the surface ratio α_l of the low-pass filter.

C. Implementation Details

Our proposed FSIOD-DFP is formulated upon Faster RCNN [63] and leverages the ResNet-101 [74] pretrained on ImageNet [75] with a feature pyramid network (FPN) [64] as the backbone. During the base training stage, the model is trained for 36 000 iterations. The learning rate initiates at 0.01 and reverses to 0.001 and 0.0001 at 24 000 and 32 000 iterations, respectively. For the subsequent training stage, we fine-tune the newly added dual-frequency prompt generator and novel detection head for 6000 iterations with a learning rate of 0.001. The balance weight λ in (2) is set to 0.1. We use the stochastic gradient descent (SGD) optimizer in both training stages. The momentum and the

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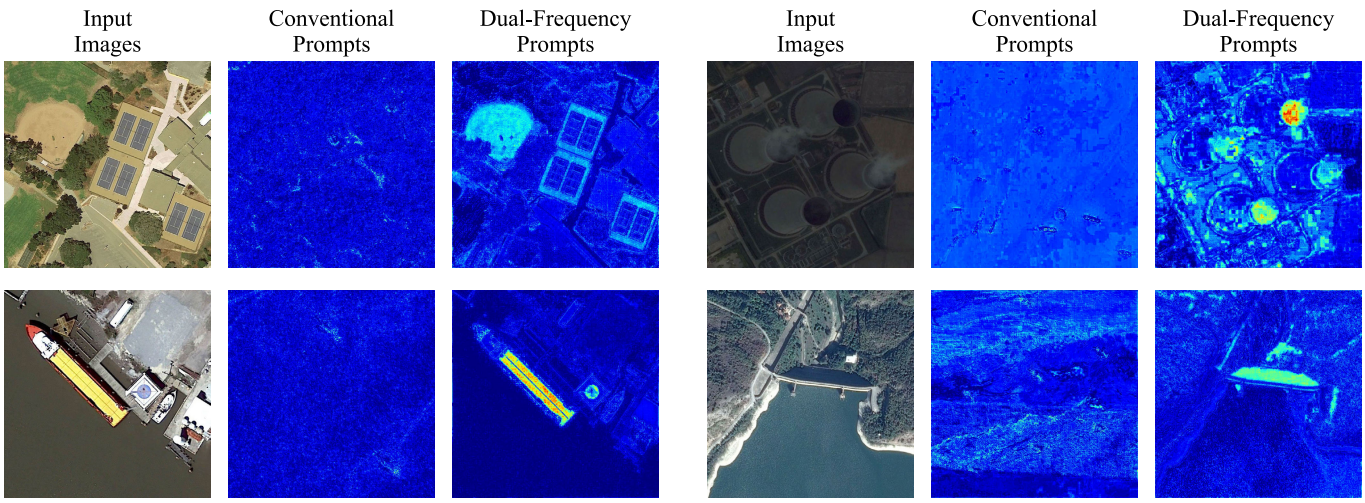


Fig. 7. Visualizations of conventional prompts and dual-frequency prompts generated by the proposed DFPG.

weight decay are 0.9 and 0.0001, respectively. The experiments are conducted on a 32-GB NVIDIA Tesla V100 GPU. We set the batch size as 4. During testing, the confidence and IoU thresholds in nonmaximum suppression (NMS) are set to 0.05 and 0.5, respectively. For DIOR and NWPU VHR-10, we utilize the original images without any modifications and select the same images as in [27] for subsequent tasks. In the FAIR1M-Airplane dataset, the images are cropped to 800×800 pixels with a 200-pixel overlap, resulting in 12 964 training patches and 5413 test patches. No data augmentation strategies are applied in either training or testing stages. Also, our approach does not replay the data of previous tasks during subsequent training. Each experiment is conducted five times, and the reported results are the averages of the five experimental results.

D. Ablation Studies

To validate the efficacy of the components in our FSIOD-DFP, we conduct ablation experiments on the first class split of the DIOR dataset under five- and ten-shot settings.

1) Ablation Study for Each Component: First, we train a Faster RCNN on the base dataset to establish a base model, achieving an mAP of 71.59% on base classes. To validate the efficacy of each module in FSIOD-DFP, we conduct experiments on each component. The results are shown in Table II. The baseline, without our proposed modules, introduces classification weights and bounding box regression weights to the detection head of the base model for the novel task. It only fine-tunes the detection head, with the remaining parameters frozen. While the baseline mitigates forgetting for the base task, the performance on the novel task is low due to the mismatch between the data of the two tasks, making it unable to extract compelling features from the novel data. Based on the baseline, adding the dual-frequency prompt generator significantly improves the performance on the novel task, indicating that the generated prompts can modify the novel data to adapt to the base model to extract compelling features and address the overfitting with limited data. However, due to the background shift, the performance on the base task decreases. Furthermore, when adding the task-decoupled detection head alone, the performance on the base task does not decrease, and the performance on the novel task is also improved compared to the baseline. When combining these two modules, the model improves the performance on the novel task, and at the same time, the performance on the base task does not decline. In addition, adding the self-regularization loss to explicitly guide the generated prompts further enhances the performance on the novel task. The experimental results show the effectiveness of the designed modules. They can learn a novel task with limited data while completely avoiding catastrophic forgetting on previous tasks.

2) Ablation Study for Different Detectors: As mentioned earlier, our method only modifies the input and output of the model, making it applicable to various backbones and object detectors. To validate the robustness of FSIOD-DFP, we conduct experiments on different backbones and detectors, as shown in Table III. For the two-stage detector Faster RCNN, we apply both convolution-based ResNet and Transformer-based Swin [46] as backbones, and all of them achieve prominent performance on both base and novel tasks. It demonstrates the robustness of our method across different backbones. In addition, we combine our proposed modules with two one-stage detectors, RetinaNet [67] and CornerNet [68]. Compared to the baseline, incorporating our proposed modules allows the model to fully retain the performance on the base task and mitigate overfitting on the novel task, significantly improving the overall performance on all tasks.

3) Ablation Study for Dual-Frequency Prompt Generator: To explore the effectiveness of the dual-frequency prompt generator, we compare the module with the conventional prompts and the prompts generated from the single-frequency component. As shown in Table IV, conventional prompts refer to constructing learnable embeddings of the same size as the input image and initializing them to zero, which are then fine-tuned on the novel task. Directly using the conventional prompts leads to decreased performance on the novel task, as introducing too many parameters and meeting overfitting. In contrast, our prompt generator generates prompts through several convolutional layers, significantly reducing

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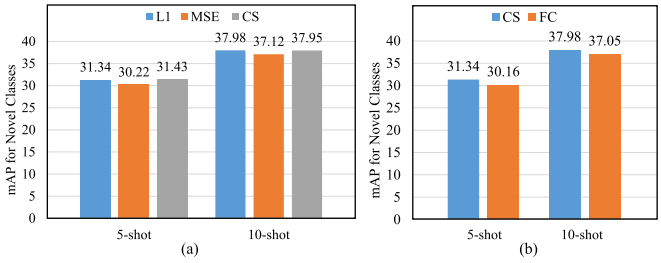


Fig. 8. (a) Ablation study for self-regularization loss. (b) Ablation study for the novel classifier in the task-decoupled detection head.

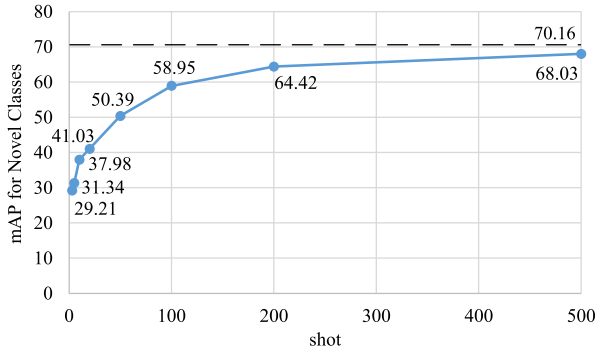


Fig. 9. Ablation study for the number of shots.

the introduced parameters and mitigating overfitting for the novel task. In addition, we demonstrate the effectiveness of applying dual-frequency components. We use a single-branch generator to generate prompts from the original input image, low-frequency component, and high-frequency component. It can be observed that using the original image improves the performance on the novel task compared to the baseline without prompts. When applying low- and high-frequency components separately, the performance on the novel task is further enhanced. Also, combining these two frequency components, i.e., dual frequency, allows for the simultaneous extraction of objects' main features from low-frequency components and boundary characteristics from high-frequency components. It filters out interference and achieves optimal overall performance on both base and novel tasks.

In addition, we conduct an ablation analysis of the hyperparameters α_h and α_l to determine the value selection, i.e., the center filtering or retention ratio in (3) and (4). Fig. 6(a) and (b), respectively, shows the performance on the novel task as α_h varies when $\alpha_l = 0.1$ and as α_l varies when $\alpha_h = 0.3$. It can be observed that the performance increases initially and then decreases with the increase in α_h and α_l. Thus, effectively filtering cluttered information and retaining valuable features can only be achieved with reasonable values. Based on the results, we set α_h and α_l to 0.3 and 0.1, respectively, in other experiments.

To explain the interpretability of dual-frequency prompts, we visually compare our generated prompts with conventional prompts, as shown in Fig. 7. The conventional prompts are unable to perceive input images and present inexplicable content. Our proposed dual-frequency prompts are generated from the low- and high-frequency components of input images and have the interpretability in modifying images. They can effectively modify the regions of the objects belonging to the

TABLE V
ABLATION STUDY FOR MULTIPLE INCREMENTAL TASKS

Incremental Settings	Shot	Base	Task 1	Task 2	Task 3	Task 4	Task 5	All
15+5	5	71.59	31.34	-	-	-	-	61.53
	10	71.59	37.98	-	-	-	-	63.19
15+3+2	5	71.59	30.78	30.35	-	-	-	61.34
	10	71.59	38.08	37.23	-	-	-	63.13
15+2+3	5	71.59	30.63	29.35	-	-	-	61.16
	10	71.59	37.86	37.52	-	-	-	63.11
15+1+1+1+1+1	5	71.59	39.34	26.81	19.35	39.58	10.10	60.45
	10	71.59	42.25	34.22	23.81	46.94	15.24	61.82

TABLE VI
ABLATION STUDY FOR CROSS-DATASET INCREMENTAL TASKS

Incremental Sequence	Shot	Base	Task 1	Task 2	All
DIOR-NWPU VHR10-FAIR1M	5	75.36	71.53	46.25	66.62
	10	75.36	75.82	47.87	68.10
DIOR-FAIR1M-NWPU VHR10	5	75.36	46.25	71.53	66.62
	10	75.36	47.87	75.82	68.10

novel classes (baseball field, chimney, and ship) and make them compatible with the base model to extract useful features for prediction. Surprisingly, the dual-frequency prompts also modify the objects belonging to the base classes that are similar to the novel classes, such as the tennis court resembling a basketball court and the dam resembling a bridge.

4) Ablation Study for Self-Regularization Loss: As described earlier, the self-regularization loss can be formulated with various constraint forms, such as $L1$, mse, and CS. We conduct experiments on the form of matching loss used in $\mathcal{L}_{\text{self}}$. The performance on the novel task is shown in Fig. 8(a). The differences between the performance with different constraint forms are small. Also, among the three forms, the performance of $L1$ loss is optimal. Thus, we choose $L1$ loss as the self-regularization loss.

5) Ablation Study for Task-Decoupled Detection Head: We compare the performance of the CS classifier and the FC classifier in the novel detection head. As shown in Fig. 8(b), for the performance on the novel task, the CS classifier is better than the FC classifier. The novel task only contains limited samples, which may increase intraclass variance. The CS classifier alleviates the influence of amplitude differences between features and only focuses on the direction information. It can reduce the intraclass variance and improve the accuracy on the novel few-shot task.

6) Ablation Study for the Number of Shots: To evaluate the performance variation of our method with the number of object instances for novel classes, we conduct experiments on the number of shots. Since the number of shots does not affect the performance of the base task, the mAP of our method on the base task is also 71.59%. Also, the mAP for the novel task is shown in Fig. 9. The black dashed line represents the result of the model trained with complete data, where the number of instances of the five novel classes is 3433, 2149, 2604, 1032, and 35 108. At this point, the mAP for novel classes is 70.16%, representing the upper bound

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TABLE VII
FSIOD PERFORMANCE ON THE DIOR DATASET

Split	Method	3-shot			5-shot			10-shot			20-shot		
		Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	FRW	49.40	7.50	38.90	49.70	12.10	40.30	49.50	18.10	41.70	50.00	22.00	43.00
	Meta-RCNN	60.62	12.02	48.47	62.01	13.09	49.78	61.55	14.07	49.68	63.21	14.45	51.02
	FsDetView	59.54	13.19	47.95	58.58	14.29	47.51	59.64	18.02	49.24	62.69	18.01	51.52
	P-CNN	47.00	18.00	39.80	48.40	22.80	42.00	50.90	27.60	45.10	52.20	29.60	46.80
	TFA w/cos	70.32	11.35	55.58	70.51	11.57	55.78	70.52	15.37	56.73	71.07	17.96	57.79
	DeRFCN	61.90	28.25	53.48	62.04	30.30	54.11	61.51	32.64	54.29	62.44	35.37	55.67
	TEMO	60.89	28.04	52.68	64.30	31.05	55.99	65.53	35.38	57.99	65.62	40.25	59.28
	iMTFA	60.13	23.34	50.93	62.51	25.20	53.18	65.81	29.35	56.70	62.43	35.22	55.63
	Incremental-DETR	65.93	27.16	56.24	66.19	30.21	57.20	69.98	36.44	61.60	70.13	37.55	61.99
	G-FSDet	68.94	27.57	58.61	69.52	30.52	59.72	69.03	37.46	61.16	69.80	39.83	62.31
	Ours	**71.59**	**29.21**	**60.99**	**71.59**	**31.34**	**61.53**	**71.59**	**37.98**	**63.19**	**71.59**	**41.03**	**63.95**
2	FRW	48.50	4.80	37.60	46.80	7.00	36.90	46.40	9.00	37.10	43.50	14.10	36.20
	Meta-RCNN	62.55	8.84	49.12	63.14	10.88	50.07	63.28	14.90	51.18	63.86	16.71	52.07
	FsDetView	58.88	10.83	46.87	60.31	9.63	47.64	61.16	13.57	49.26	61.16	14.76	49.56
	P-CNN	48.90	14.50	40.30	49.10	14.90	40.60	52.50	18.90	44.10	51.60	22.80	44.40
	TFA w/cos	**70.75**	5.77	54.51	**70.79**	8.19	55.14	**69.93**	8.71	54.63	**70.02**	12.18	55.56
	DeRFCN	61.41	14.55	49.70	61.79	16.45	50.45	61.42	18.40	50.69	62.79	21.13	52.38
	TEMO	61.39	15.57	49.94	65.23	17.98	53.42	62.62	21.04	52.23	61.18	23.04	51.65
	iMTFA	59.97	13.13	48.26	60.26	14.92	48.93	60.35	19.37	50.11	61.32	23.32	51.82
	Incremental-DETR	67.29	12.57	53.61	66.98	15.04	54.00	67.15	20.10	55.39	68.03	20.17	56.07
	G-FSDet	69.20	14.13	55.43	69.25	15.84	55.87	68.71	20.70	56.70	68.18	22.69	56.86
	Ours	69.91	**17.94**	**56.92**	69.91	**20.19**	**57.48**	69.91	**21.85**	**57.90**	69.91	**23.96**	**58.42**
3	FRW	45.50	7.80	36.10	47.90	13.70	39.30	44.50	13.80	36.80	43.50	18.50	37.30
	Meta-RCNN	61.93	9.10	48.72	63.44	12.29	50.66	62.57	11.96	49.92	65.53	16.14	53.18
	FsDetView	61.00	7.49	47.63	61.33	12.61	49.15	61.94	11.49	49.32	65.17	17.02	53.14
	P-CNN	49.50	16.50	41.30	49.90	18.80	42.10	52.10	23.30	44.90	53.10	28.80	47.00
	TFA w/cos	71.95	8.36	56.05	71.64	10.13	56.26	**72.56**	10.75	57.11	**73.13**	17.99	59.35
	DeRFCN	63.56	15.78	51.47	63.25	18.73	52.11	64.55	20.43	53.52	64.56	25.13	54.71
	TEMO	65.88	16.02	53.42	65.76	22.83	55.03	67.75	24.68	56.98	67.74	30.57	58.45
	iMTFA	63.78	15.93	51.82	64.33	20.15	53.29	65.55	23.46	55.03	65.93	27.85	56.41
	Incremental-DETR	68.33	16.35	55.34	68.84	22.78	57.33	69.35	25.03	58.27	67.83	31.34	58.78
	G-FSDet	71.10	16.03	57.34	70.18	23.25	58.43	71.08	26.24	59.87	71.26	**32.05**	61.46
	Ours	**72.50**	**16.97**	**58.62**	**72.50**	**25.37**	**60.72**	72.50	**29.51**	**61.76**	72.50	32.00	**62.38**
4	FRW	48.20	3.70	37.10	48.50	6.80	38.10	45.70	7.20	36.10	44.40	12.20	36.40
	Meta-RCNN	61.73	13.94	49.78	62.60	15.84	50.91	62.23	15.07	50.44	63.24	18.17	51.98
	FsDetView	58.90	14.28	47.75	58.97	15.95	48.22	60.37	15.37	49.12	60.89	16.96	49.91
	P-CNN	49.80	15.20	41.20	49.90	17.50	41.80	51.70	18.90	43.50	52.30	25.70	45.70
	TFA w//cos	68.57	10.42	54.03	68.85	14.29	55.21	68.58	14.35	55.03	68.86	12.01	54.65
	DeRFCN	59.81	10.83	47.50	59.71	18.62	49.44	59.54	21.61	49.99	60.09	27.61	51.97
	TEMO	66.35	17.13	54.05	66.30	21.28	55.05	66.82	26.77	56.81	67.24	32.08	58.45
	iMTFA	63.39	15.44	51.40	60.49	19.75	50.31	64.97	24.32	54.81	62.03	31.00	54.27
	Incremental-DETR	67.05	14.48	53.91	66.31	20.55	54.87	64.38	25.65	54.70	66.37	28.49	56.90
	G-FSDet	69.01	16.74	55.95	67.96	21.03	56.30	68.55	25.84	57.87	67.73	31.78	58.75
	Ours	**70.32**	**18.03**	**57.24**	**70.32**	**21.82**	**58.20**	**70.32**	**27.74**	**59.67**	**70.32**	**34.58**	**61.39**

of the performance on the few-shot novel task. We conduct experiments with shots of 3, 5, 10, 20, 50, 100, 200, and 500. As the shot increases, the performance on the novel classes gradually improves, approaching the performance upper bound of fully supervised training with complete data. Moreover, when the number of shots is small, the performance improves rapidly with the increase of shots. For example, as the number of shots increases from 3 to 50, the mAP increases from 29.21% to 50.39%. Continuing to increase shot, the upward trend of the mAP for novel classes slows down. When the shot is 500, the number of object instances used for training is only about 5% of the total instances, and the mAP for novel classes only decreases by 2.13% compared with fully supervised training on complete data, reaching 68.03%. It indicates that our method effectively reduces the number of annotated instances for learning novel tasks, which is suitable for few-shot incremental scenarios.

7) Ablation Study for Complex Incremental Tasks: In addition to the incremental setting with only one novel task, i.e., $15 + 5$ (a base task with 15 categories and a novel task with five categories), we also conduct experiments in complex few-shot incremental settings, including two scenarios of adding two novel tasks ($15 + 3 + 2$ and $15 + 2 + 3$) and one scenario of adding five novel tasks ($15 + 1 + 1 + 1 + 1 + 1$).

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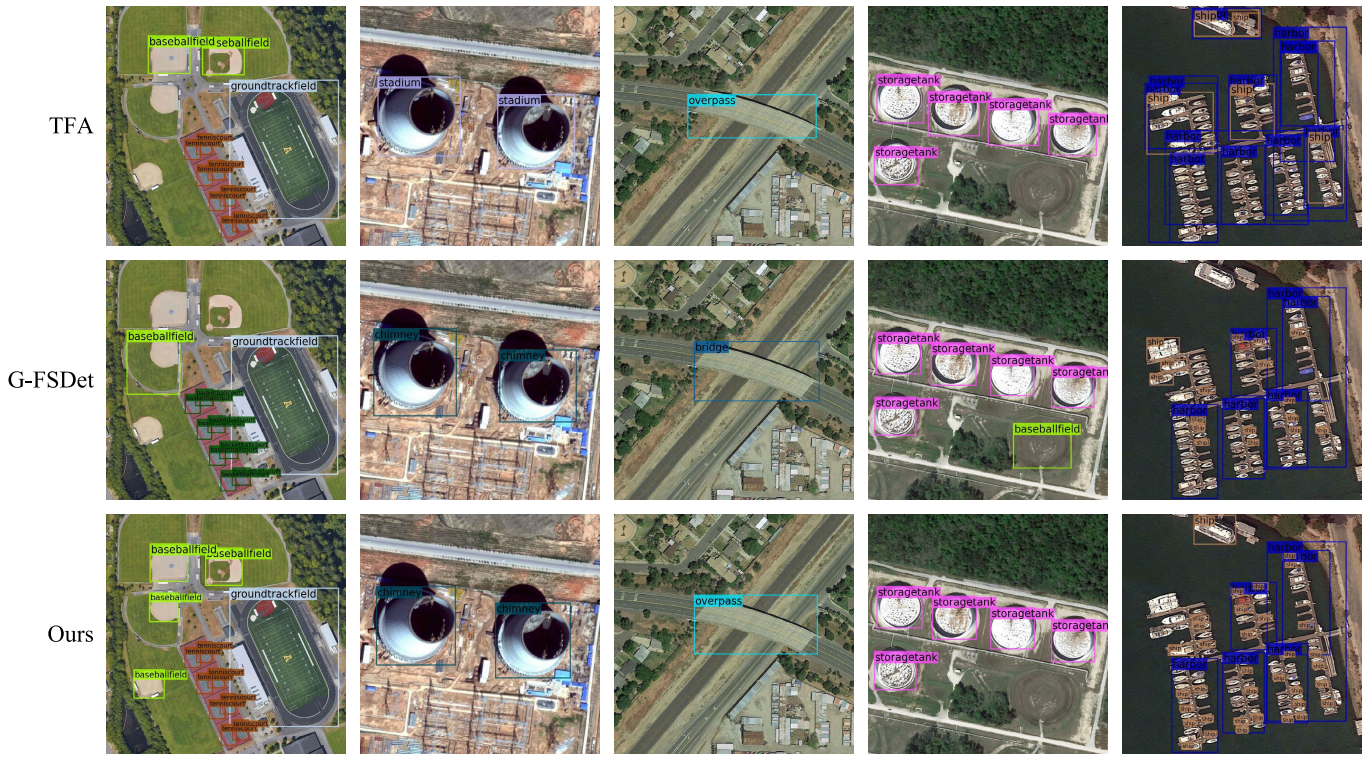


Fig. 10. Visualization comparisons between TFA, G-FSDet, and our method on Split 1 of the DIOR dataset in the ten-shot setting.



Fig. 11. Visualizations of few-shot training images for Split 1 of the DIOR dataset in the ten-shot setting. Each novel class consists of ten annotated object instances.

The categories of the novel tasks are arranged in alphabetical order. The results are shown in Table V. Our method exhibits only a slight decrease in overall performance when faced with complex scenarios of multiple incremental tasks. For the two scenarios with two novel tasks, the overall performance on all tasks decreases by an average of 0.28% and 0.07% in five-shot and ten-shot, respectively. When adding five novel tasks, the performance decreases by 1.08% and 1.37% in five-shot and ten-shot, respectively. It indicates that FSIOD-DFP is capable of handling the scenarios with multiple incremental tasks.

Furthermore, we conduct experiments in a more challenging incremental scene, i.e., cross-dataset few-shot incremental tasks. The DIOR dataset serves as the base task. The NWPU VHR-10 and FAIR1M-Airplane datasets are applied as the novel tasks. A unified model is trained and evaluated on these three datasets. As shown in Table VI, even in the challenging scenario of cross-dataset FSIL, our FSIOD-DFP still achieves splendid performance. Besides, the performance of FSIOD-DFP is independent of the learning order of subsequent incremental tasks, which only depends on the quality of the base model. Thus, training on a well-annotated base dataset to obtain the base model will enhance the performance of our FSIOD-DFP on all tasks.

E. Comparisons With State-of-the-Art Methods

To showcase the effectiveness of our FSIOD-DFP, we conduct comparisons with state-of-the-art methods across three datasets, including DIOR, NWPU VHR-10, and FAIR1M-Airplane. For each dataset, an individual model is built and trained. The compared methods include few-shot object detection methods (FRW [14], Meta-RCNN [15], FsDetView [16], P-CNN [17], TFA [18], DeRFCN [19], and TEMO [20]) and FSIOD methods (iMTFA [41], Incremental-DETR [42], and G-FSDet [27]). Some methods are designed for natural scene images [15], [16], [18], [19], [41], [42], and some methods are specifically designed for remote sensing images [14], [17], [20], [27].

1) Experiments on DIOR: Table VII presents the comparison results between our proposed method and state-of-the-art methods on the DIOR dataset, reflecting the performance on the base task and the novel task, including three-, five-, ten-, and 20-shot settings of four class splits. It can be observed that few-shot object detection methods only consider the overfitting of the novel task but result in a performance decline on the base task after training on the novel task, i.e., catastrophic forgetting. Among them, TFA achieves the best performance on the base task, but its performance on the novel task is poor due to freezing most of the network. In contrast, FSIOD

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TABLE VIII
FSIOD PERFORMANCE ON THE NWPU VHR-10 DATASET

Split	Method	3-shot			5-shot			10-shot			20-shot		
		Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	FRW	83.13	15.35	62.80	82.78	16.24	62.82	83.89	24.00	65.92	82.80	27.16	66.11
	Meta-RCNN	87.00	20.51	67.05	85.74	21.77	66.55	87.01	26.98	69.00	87.29	28.24	69.57
	FsDetView	87.68	24.56	68.75	87.77	29.55	70.31	87.75	31.77	70.96	87.83	32.73	71.30
	P-CNN	82.84	41.80	70.53	82.89	49.17	72.79	83.05	63.29	78.11	83.59	66.83	78.55
	TFA w/cos	89.35	8.80	65.19	89.60	9.49	64.65	89.95	9.26	65.74	89.62	10.83	65.98
	DeRFCN	87.22	37.90	73.33	86.87	46.08	74.63	87.29	62.95	79.99	87.26	64.61	80.46
	TEMO	86.35	**57.57**	77.72	85.29	66.77	79.73	87.11	**75.07**	83.49	89.22	75.89	85.22
	iMTFA	86.78	43.20	73.71	86.03	49.95	75.21	87.24	70.01	82.07	87.93	72.78	83.39
	Incremental-DETR	87.16	50.93	76.29	87.05	53.18	76.89	87.89	69.94	82.51	88.35	73.18	83.80
	G-FSDet	89.11	49.05	77.01	88.37	56.10	78.64	88.40	71.82	83.43	89.73	75.41	85.44
	Ours	**90.56**	55.33	**79.99**	**90.56**	**67.20**	**83.55**	**90.56**	74.95	**85.87**	**90.56**	**77.10**	**86.52**
2	FRW	85.34	28.61	68.32	85.75	31.45	69.45	85.49	32.53	69.60	86.58	33.93	70.79
	Meta-RCNN	86.86	21.41	67.23	87.38	35.34	71.77	87.56	37.14	72.43	87.26	39.47	72.92
	FsDetView	88.11	39.01	73.38	89.34	40.31	74.63	89.34	45.09	76.07	89.31	46.28	76.40
	P-CNN	81.03	39.32	68.52	81.18	46.10	70.70	80.93	55.90	73.41	81.21	58.37	75.50
	TFA w/cos	90.14	11.14	66.44	91.19	12.46	67.57	90.79	11.35	66.96	90.37	11.56	66.73
	DeRFCN	88.47	39.19	73.69	88.17	45.56	75.29	88.25	54.05	77.99	88.05	57.38	78.83
	TEMO	88.25	50.82	77.02	88.31	59.31	79.61	89.07	64.52	81.71	87.93	73.35	83.56
	iMTFA	87.36	46.29	75.04	88.05	54.34	77.94	88.03	62.56	80.39	88.32	66.37	81.74
	Incremental-DETR	89.12	47.35	76.59	89.37	57.24	79.73	88.87	65.63	81.90	89.25	70.58	83.65
	G-FSDet	89.99	50.09	78.02	90.52	58.75	80.99	89.23	67.00	82.56	90.61	75.86	86.13
	Ours	**91.82**	**52.15**	**79.92**	**91.82**	**60.98**	**82.57**	**91.82**	**70.94**	**85.56**	**91.82**	**78.36**	**87.78**

TABLE IX
FSIOD PERFORMANCE ON THE FAIR1M-AIRPLANE DATASET

Split	Method	2-shot			3-shot			5-shot			10-shot		
		Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	FRW	39.12	22.35	34.55	40.04	23.50	35.53	42.36	25.25	37.69	45.83	30.45	41.64
	Meta-RCNN	41.98	24.57	37.23	47.32	27.80	42.00	49.45	28.80	43.82	49.31	35.73	45.61
	FsDetView	43.92	25.30	38.84	50.23	29.70	44.63	51.04	29.82	45.25	50.67	34.60	46.29
	P-CNN	44.57	30.95	40.86	49.78	32.58	45.09	51.27	36.34	47.20	53.35	39.98	49.70
	TFA w/cos	50.46	28.93	44.59	52.44	31.13	46.63	54.39	30.67	47.92	54.82	30.37	48.15
	DeRFCN	43.57	30.12	39.90	43.14	33.25	40.44	45.25	35.56	42.61	49.31	37.28	46.03
	TEMO	49.98	30.60	44.69	50.25	32.80	45.49	51.39	37.37	47.57	52.31	40.83	49.18
	iMTFA	45.86	29.87	41.50	47.22	29.90	42.49	49.24	33.49	44.94	50.31	39.15	47.27
	Incremental-DETR	50.32	31.02	45.06	50.10	32.17	45.21	50.83	33.86	46.20	51.56	39.82	48.36
	G-FSDet	51.15	30.34	45.47	51.93	31.98	46.49	52.37	35.40	47.74	52.04	41.04	49.04
	Ours	**55.37**	**32.24**	**49.06**	**55.37**	**33.85**	**59.50**	**55.37**	**40.18**	**51.23**	**55.37**	**42.55**	**51.87**

methods can alleviate catastrophic forgetting for the base task when learning the novel task. G-FSDet, designed for remote sensing images, outperforms other methods on both novel and base tasks. It is worth noting that these methods need to replay a few data of previous tasks when subsequent training, and they only mitigate catastrophic forgetting. Our method does not require the data of previous tasks when learning novel tasks and can completely avoid catastrophic forgetting. Thus, the performance on the base task of our method does not decline after learning the novel task and does not change with the variation in the amount of the data (shot) of the novel task. Moreover, our method achieves the best performance in all settings for all class splits. It demonstrates the effectiveness and reliability of FSIOD-DFP for remote sensing images.

Fig. 10 presents a qualitative analysis of the results of our FSIOD-DFP compared to TFA with excellent performance on the base task and the state-of-the-art method G-FSDet. Fig. 11 presents the few-shot training images used during the subsequent training, which is consistent with other methods. Due to freezing most of the network, TFA experiences a less performance decline on the base task but struggles with learning the novel task, failing to detect objects of novel classes, such as “baseball field” and “chimney” shown in Fig. 10. Moreover, G-FSDet learns the novel task effectively but exhibits slight catastrophic forgetting for base classes, such as “tennis court” and “overpass” shown in Fig. 10. In comparison, our method can completely eliminate catastrophic forgetting and effectively learn the novel task, which performs well on all tasks. Besides, for the novel class “ship” in Fig. 10, due to its small size, dense arrangement, and limited diversity in few-shot training samples shown in Fig. 11, all three methods face some missing detection. Compared to TFA and G-FSDet, our proposed method has the fewest missing detection. In general, experimental results on the DIOR

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dataset show that our method can learn novel classes without catastrophic forgetting for base classes, showing excellent prediction results in both novel and base classes.

2) Experiments on NWPU VHR-10: We further compare FSIOD-DFP with state-of-the-art methods on the two splits of the NWPU VHR-10 dataset. As shown in Table VIII, few-shot object detection methods still suffer from catastrophic forgetting and cannot achieve excellent performance on both base and novel tasks simultaneously. Among existing FSIOD methods, G-FSDet emerges as the top performer. Also, our method still completely avoids forgetting on the base task and achieves state-of-the-art performance on both base and novel tasks. These results again indicate that our method exhibits good generalization for different datasets.

3) Experiments on FAIR1M-Airplane: In complement to the two datasets mentioned above for conventional remote sensing object detection, the adaptability of our proposed FSIOD-DFP on fine-grained remote sensing object detection is evaluated. We conduct experiments on the FAIR1M-Airplane dataset and compare our method with relevant works. As shown in Table IX, in comparison to other methods, our FSIOD-DFP continues to attain state-of-the-art performance on all tasks, underscoring its effectiveness in the few-shot incremental scenario of remote sensing fine-grained object detection.

V. Conclusion

This article proposes a novel FSIOD method for remote sensing optical images based on the rich information and frequency characteristics contained in optical images. The proposed method can completely eliminate catastrophic forgetting while mitigating overfitting. It does not require replaying previous data when learning new tasks, and the performance is independent of the learning order of novel tasks. First, a dual-frequency prompt generator is proposed to address the parameter competition issue. It decouples frequencies of input images to filter out useless information for generating prompts that modify the data to adapt to the base model. Therefore, the base model can be fully frozen when learning new tasks to retain the performance of previous tasks. Besides, we introduce a self-regularization loss to explicitly guide the generated prompts to fully leverage the knowledge of the base model. Furthermore, a task-decoupled detection head is designed to address the background shift problem. It makes the background of new tasks completely independent of previous tasks. Also, a CS classifier is applied in the novel detection head to mitigate the large intraclass variance in limited data of new tasks. The experimental results on three remote sensing object detection datasets demonstrate the effectiveness of our proposed method, which achieves state-of-the-art overall performance on all tasks. In the future, we will focus on overfitting for new tasks with limited data, aiming to further improve the performance of new few-shot tasks on top of completely eliminating catastrophic forgetting. In addition, as another important branch of remote sensing images, synthetic aperture radar (SAR) images can overcome the influence of weather conditions and enable all-weather and all-day observation of the Earth. Considering the unique imaging and scattering properties of SAR images [76], [77], we will research object detection models designed for SAR images [78], [79] and combine the SAR characteristics to construct FSIOD models tailored for SAR images.

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